

# Manual

## Graphic Energy Monitor (MOC)

### EMG-GEM 8.1

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# 1 Energy Management – Energy Monitor

## Purpose

Evaluation function for energy data. Hierarchically selected, tabular representation of the power meters in the company. Representation of the logical power meters assigned to the machines and systems and display of the consumption values and comparison values. Display of the meter hierarchy.

You use the function package when:

- You wish to have an overview of the latest development in energy consumptions since the last invoice.

## Integration

The evaluations require the function package EMG-MGM for data collection as the basis.

## Features

- Display of the current energy meters and power meters in a dynamic tree-like structure.
- Display of the machines and other logistics objects in the structure of the meters and their connections
- Dynamic forming of the object hierarchy, allowing for a wide range of selection criteria such as meters, machines, evaluation groups and other logistics objects.
- Display of the meters, consumptions at the meters, limits and limit value infringements as well as implausible configurations and values in the graphic nodes.
- Energy balance: Display of the energy differences in the meter hierarchy
- Drill-down mechanism in the branches of the tree representation.

## 2 Consumption/Energy Monitor

### Overview

|                        |                                                                        |
|------------------------|------------------------------------------------------------------------|
| Menu                   | Production Facility Management → Resource analysis<br>→ Energy monitor |
| Transaction code       | conmon                                                                 |
| Function authorization | conmon                                                                 |

This document provides a description of the "Consumption/ energy monitor" application in the Manufacturing Operation Center (MOC).

### Purpose

The energy monitor lists the energy counters (resources). The energy monitor shows the current status for these resources, allowing a comparison with comparative resources. This makes it possible to display an energy balance. The system always shows the counted quantities since the last inventory, i.e. since the counter was reset.

You can also display the balance in a graphic. The same applies to the counter hierarchy defined by the BOM including the meter readings of the selected resource.

When you select data, you can use the defined counter hierarchies in tree-like structure to filter the resources in the table multiple times.

### Integration

This application is connected with the following applications:

- Consumption analysis to chronologically distribute consumption values
- Consumption statement to analyze the totals and to reset data.
- You can also visualize the monitor values in the graphic machinery due to the integration of the counters (resources).

### Selection criteria

#### Resource type

Type of resource.

Workplaces and machines always have the resource type MNR. But you can configure individual resource types for the other resources. Predefined resource types include:

DNC      NC/DNC program

|     |                                     |
|-----|-------------------------------------|
| DOC | Document                            |
| ENT | Removal device                      |
| MNR | Workplace/Machine                   |
| PAC | Packaging, transportation container |
| PRM | Test and measuring equipment        |
| PER | Production staff / general          |
| PRU | Setup staff                         |
| TEM | Tempering equipment                 |
| VOR | Device                              |
| WNR | Tool                                |

**Resource from ... to ...**

This selection criterion refers to the resource. You can also run a search using wildcards (placeholders \* and ?).

**Designation (name)**

Name of the resource.

**Cost center**

Cost center of the resource.

**Responsibility area**

Responsibility area to which the resource is assigned.

**Resource family**

The resource family to which the resource is assigned.

**Storage location**

Regular storage location of the resource.

**User fields**

MD user fields 1- 6 of the resource.

**Reference**

The resource's internal ID.

**Field descriptions****Resource type**

Resource type of the resource. By default, the HYDRA system includes some default resource types. You can configure additional resource types in HYDRA.

**Resource**

Resource as a reference point for comparisons.

**Comparison resource 1**

Resource whose values are to be compared with those of the reference point.

**Current value, absolute (comparison resource 1)**

Absolute difference between the values of the reference resource and the comparison resource.

**Current value, in percent (comparison resource 1)**

Relative difference between the reference resource values and the comparison resource values.

**Comparison resource 2**

Resource whose values are to be compared with those of the reference point.

**Current value, absolute (comparison resource 2)**

Absolute difference between the values of the reference resource and the comparison resource.

**Current value, in percent (comparison resource 2)**

Relative difference between the reference resource values and the comparison resource values.

**Reset time**

Date on which the resource counters were reset.

**Setting a reset time**

The reset date depends on the entered records.

If you select a reset date that coincides with a resource record period, the system sets the reset time to the start time of this resource record.



If you select a reset date that does not coincide with a resource record period, the system uses the start time of the next resource record as the reset time.

The system uses the reset time you enter if:

- you select a reset date that does not coincide with a resource record period and
- there is no record that starts at a later point in time.